100 mm (3.94 in) bore × 80 mm (3.15 in) rod Cylinder Force Data

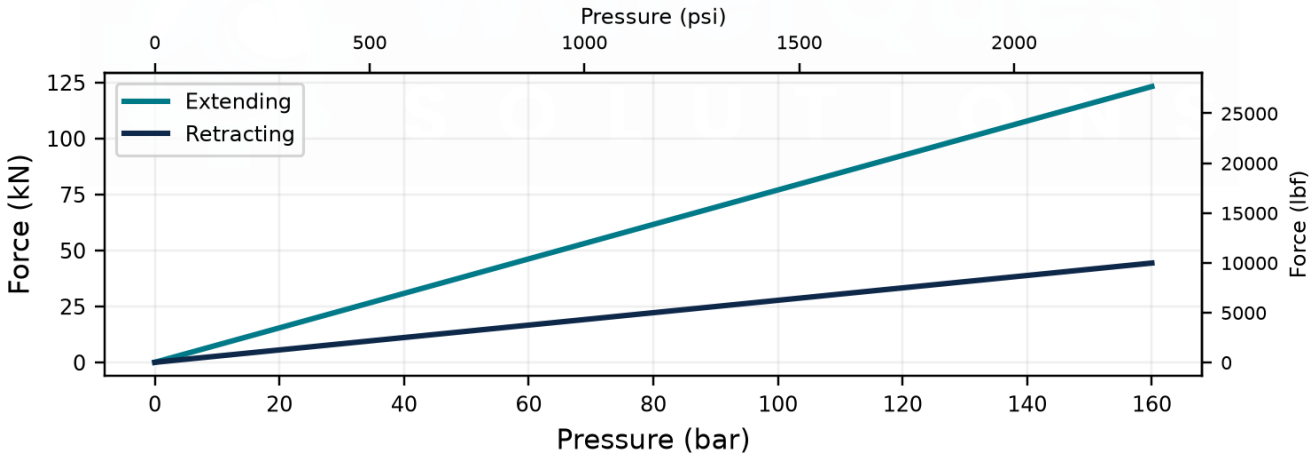
This geometry has a retraction area equal to 36.0% of its full piston area. Compare push and pull force at the same pressure, then check stroke, buckling, mounting and return-side pressure for the actual load.

Complete force output data across all operating pressures from 0 to 160 bar (0 to 2,321 psi). All values calculated with 0.98 efficiency factor.

At a glance

Parameter / value	Parameter / value
Piston Area 78.54 cm² (12.174 in²)	Ring Area 28.27 cm² (4.382 in²)
Max Extending Force 123.15 kN (27,685 lbf)	Max Retracting Force 44.33 kN (9,966 lbf)

Push and pull force



Plotted from the webpage force table, including its stated 0.98 efficiency factor.

Selection basis

Match the selected configuration to the working water, pressure, flow, temperature and duty. Maximum ratings and separate curve points do not establish one simultaneous operating point. Confirm installation and supplied configuration with WaterQuest Solutions.

Technical data

Specifications and selection data

Pressure	Force Extending (kN)	Force Retracting (kN)
0 bar (0 psi)	0.00 kN (0 lbf)	0.00 kN (0 lbf)
10 bar (145 psi)	7.70 kN (1,731 lbf)	2.77 kN (623 lbf)
20 bar (290 psi)	15.39 kN (3,460 lbf)	5.54 kN (1,245 lbf)
30 bar (435 psi)	23.09 kN (5,191 lbf)	8.31 kN (1,868 lbf)
40 bar (580 psi)	30.79 kN (6,922 lbf)	11.08 kN (2,491 lbf)
50 bar (725 psi)	38.48 kN (8,651 lbf)	13.85 kN (3,114 lbf)
60 bar (870 psi)	46.18 kN (10,382 lbf)	16.62 kN (3,736 lbf)
70 bar (1,015 psi)	53.88 kN (12,113 lbf)	19.39 kN (4,359 lbf)
80 bar (1,160 psi)	61.58 kN (13,844 lbf)	22.16 kN (4,982 lbf)
90 bar (1,305 psi)	69.27 kN (15,573 lbf)	24.93 kN (5,604 lbf)
100 bar (1,450 psi)	76.97 kN (17,304 lbf)	27.70 kN (6,227 lbf)
110 bar (1,595 psi)	84.67 kN (19,035 lbf)	30.48 kN (6,852 lbf)
120 bar (1,740 psi)	92.36 kN (20,763 lbf)	33.25 kN (7,475 lbf)
130 bar (1,885 psi)	100.06 kN (22,494 lbf)	36.02 kN (8,098 lbf)
140 bar (2,031 psi)	107.76 kN (24,225 lbf)	38.79 kN (8,720 lbf)
150 bar (2,176 psi)	115.45 kN (25,954 lbf)	41.56 kN (9,343 lbf)
160 bar (2,321 psi)	123.15 kN (27,685 lbf)	44.33 kN (9,966 lbf)